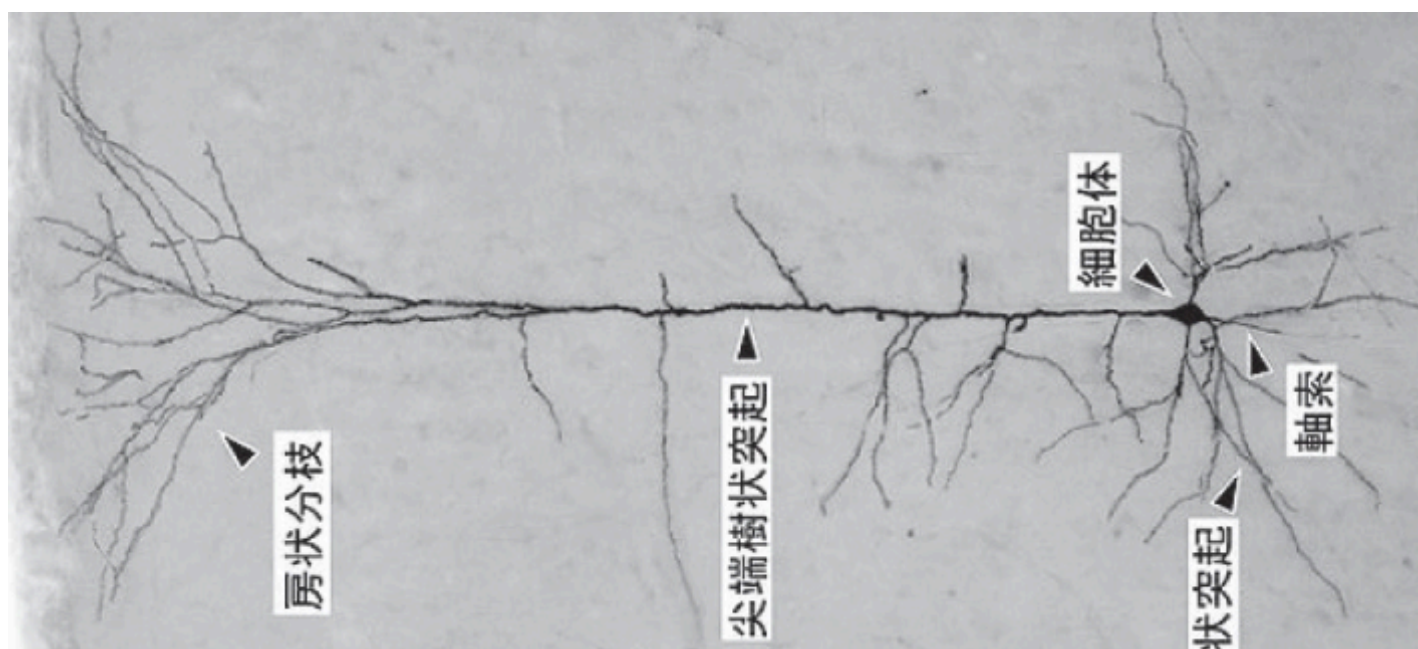


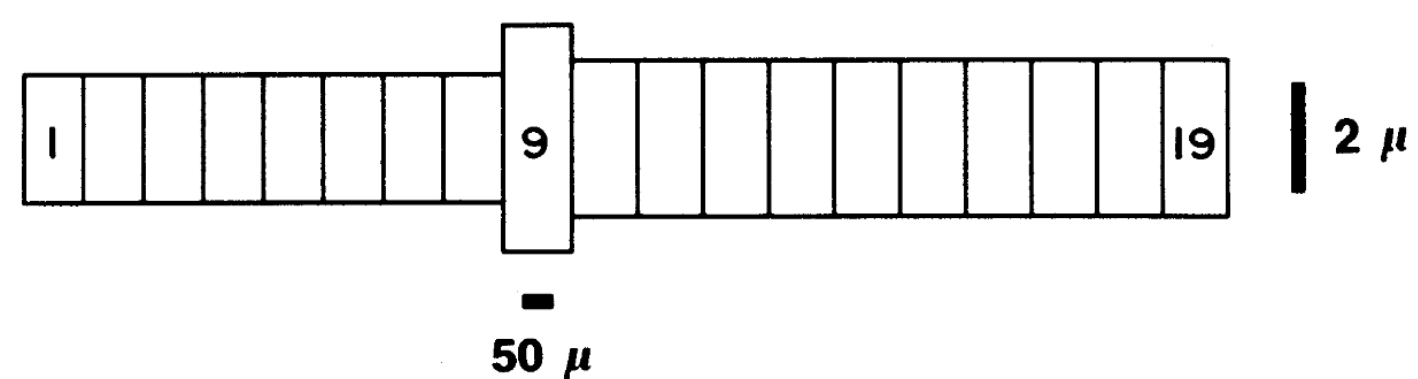
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

Haruki Munechika

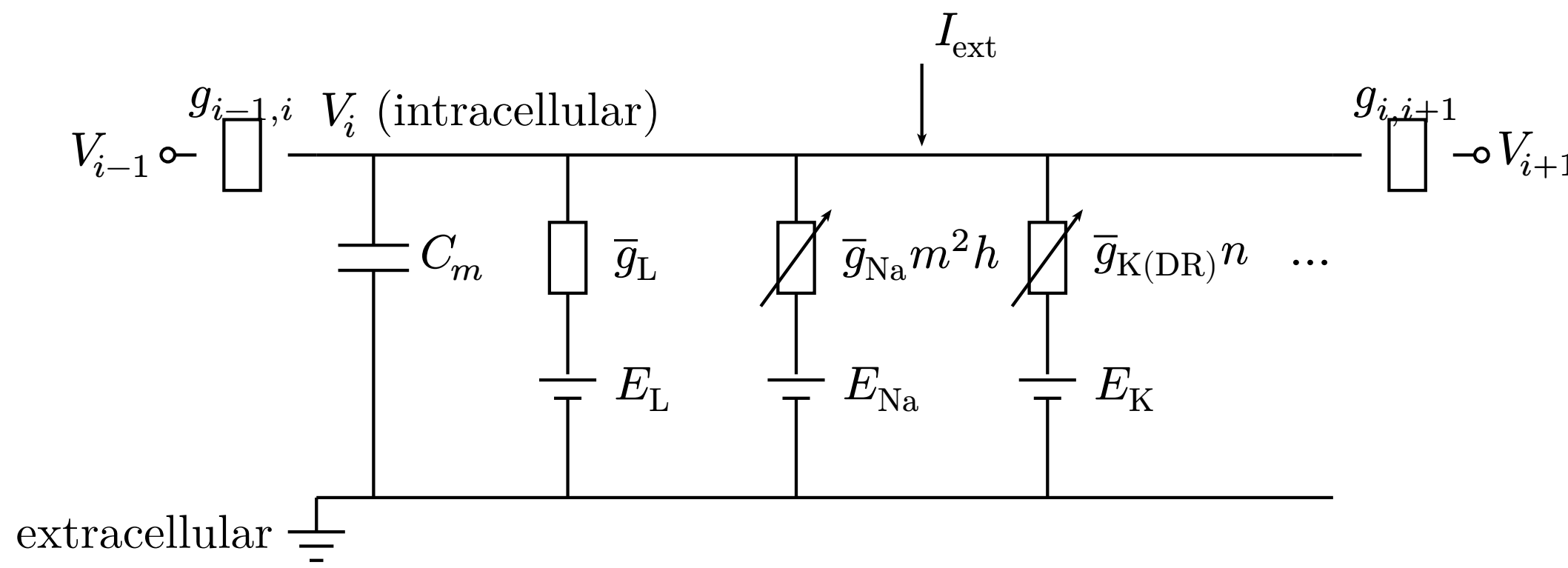
Introduction



CA3 pyramidal neuron [1]



Modelled as **19 compartments** [2]



Equivalent circuit of compartment i

Each compartment: capacitance, **variable** ionic conductances, axial coupling $g_{i,i\pm 1}$.

Goal

Test whether **complex gate dynamics** remain tractable to a **simple regression** (SINDy) after a coordinate change — and thereby reproduce the membrane-potential response with **fewer state variables**.

Conductances depend on **gate variables**, ODEs with rate functions α, β :

$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$

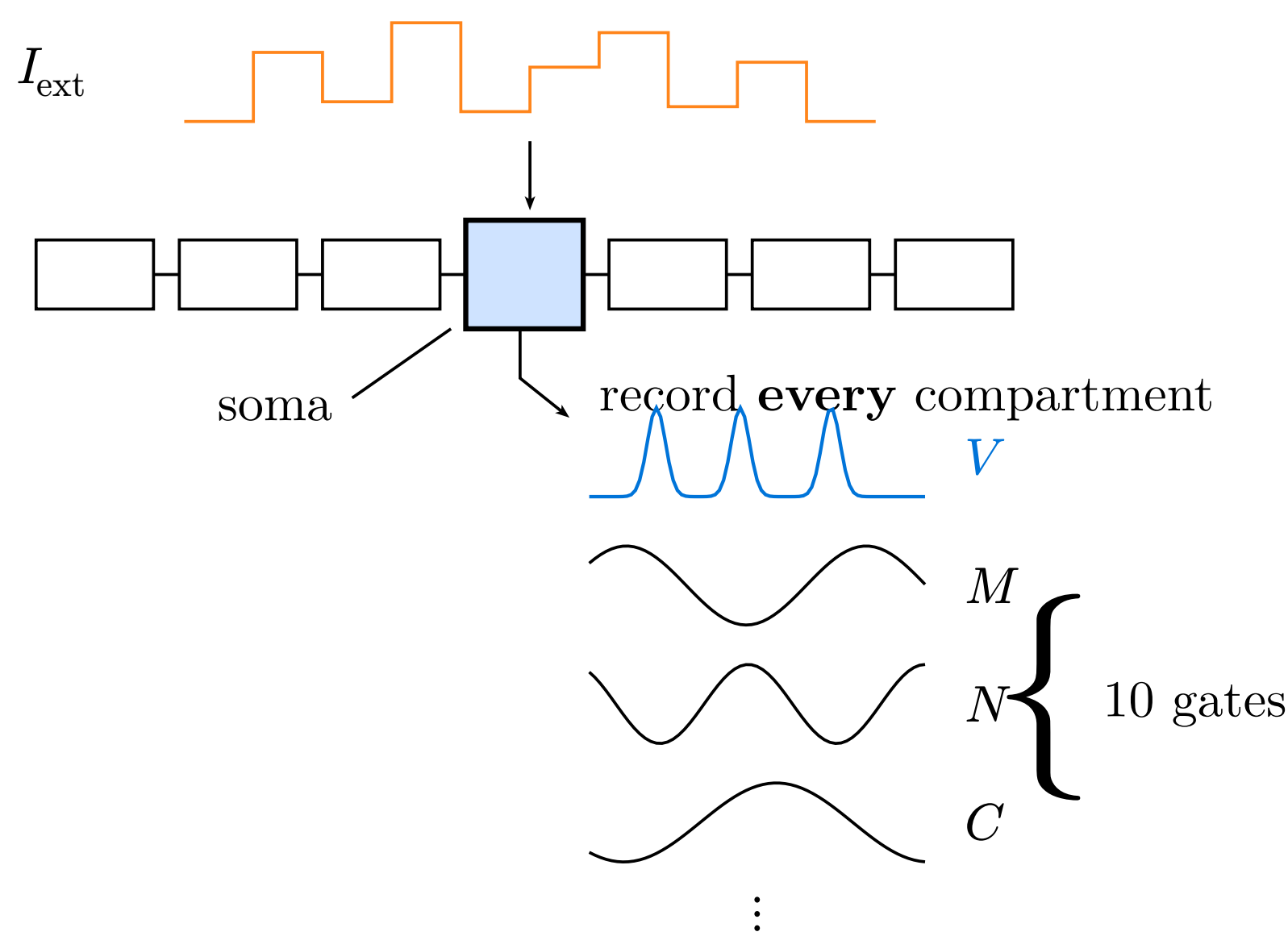
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$

→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

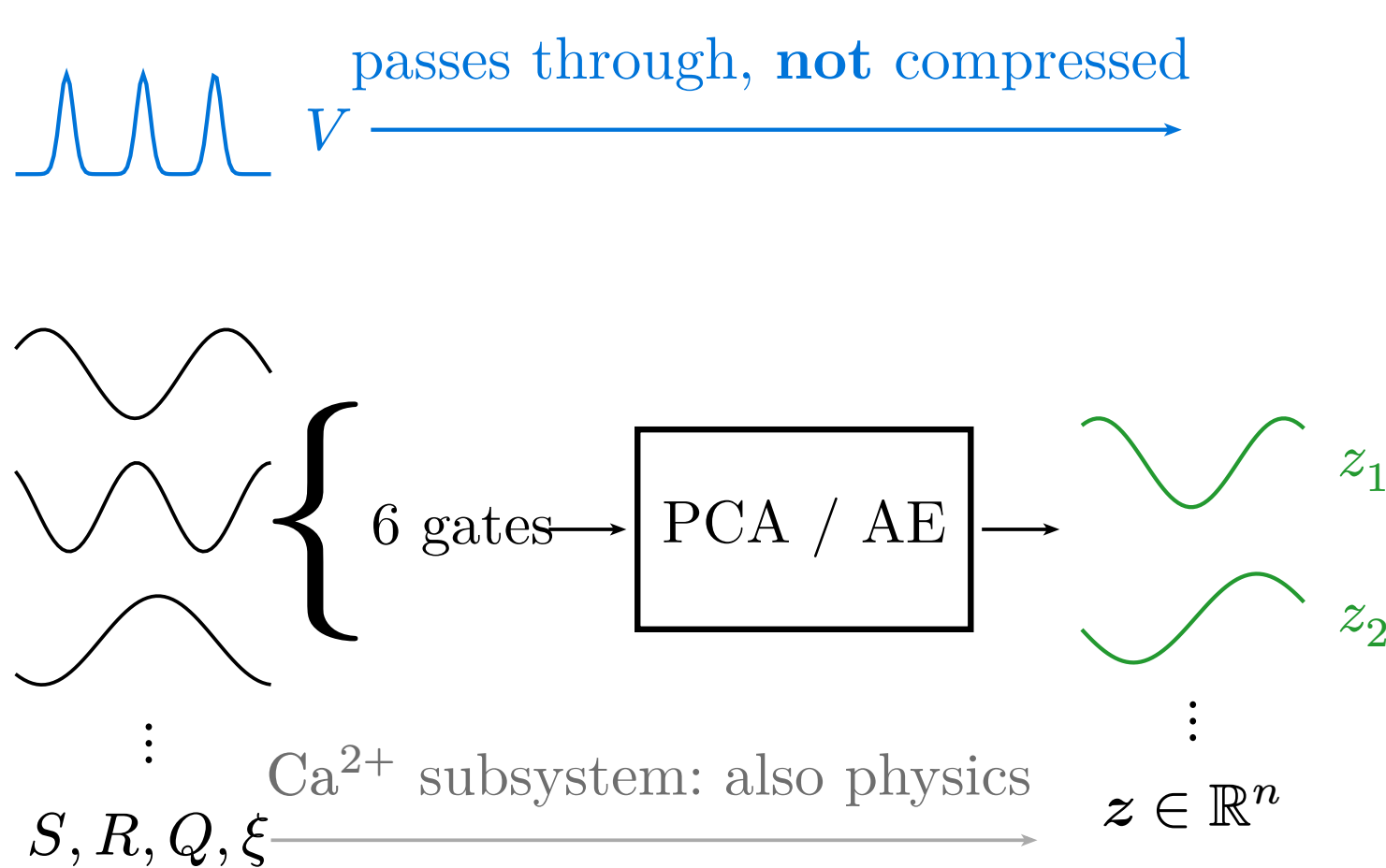
Methods

① Collect the training data



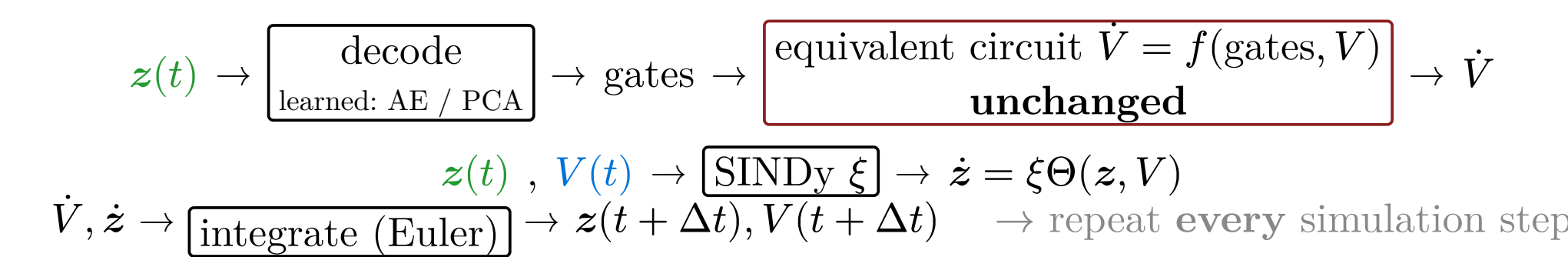
Inject a **random pulse train** at the soma; record V and **10 gates** for **all 19 compartments**.

② Compress *only* the gates

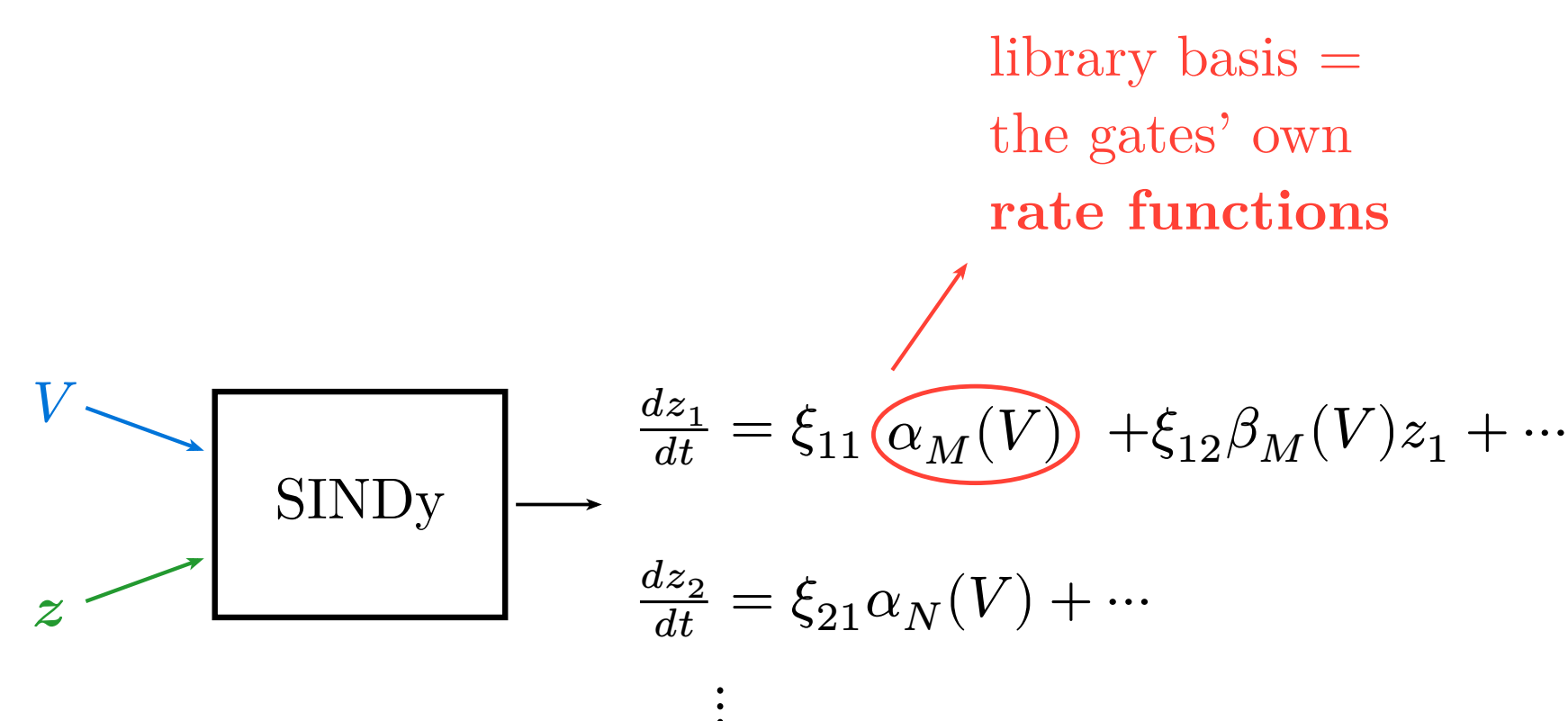


6 voltage-dependent gates ride a **low-dimensional manifold** → $z \in \mathbb{R}^n$, $n = 5$. V and Ca^{2+} **stay uncompressed**.

④ Simulate: decoder-in-the-loop



③ Identify the latent ODEs

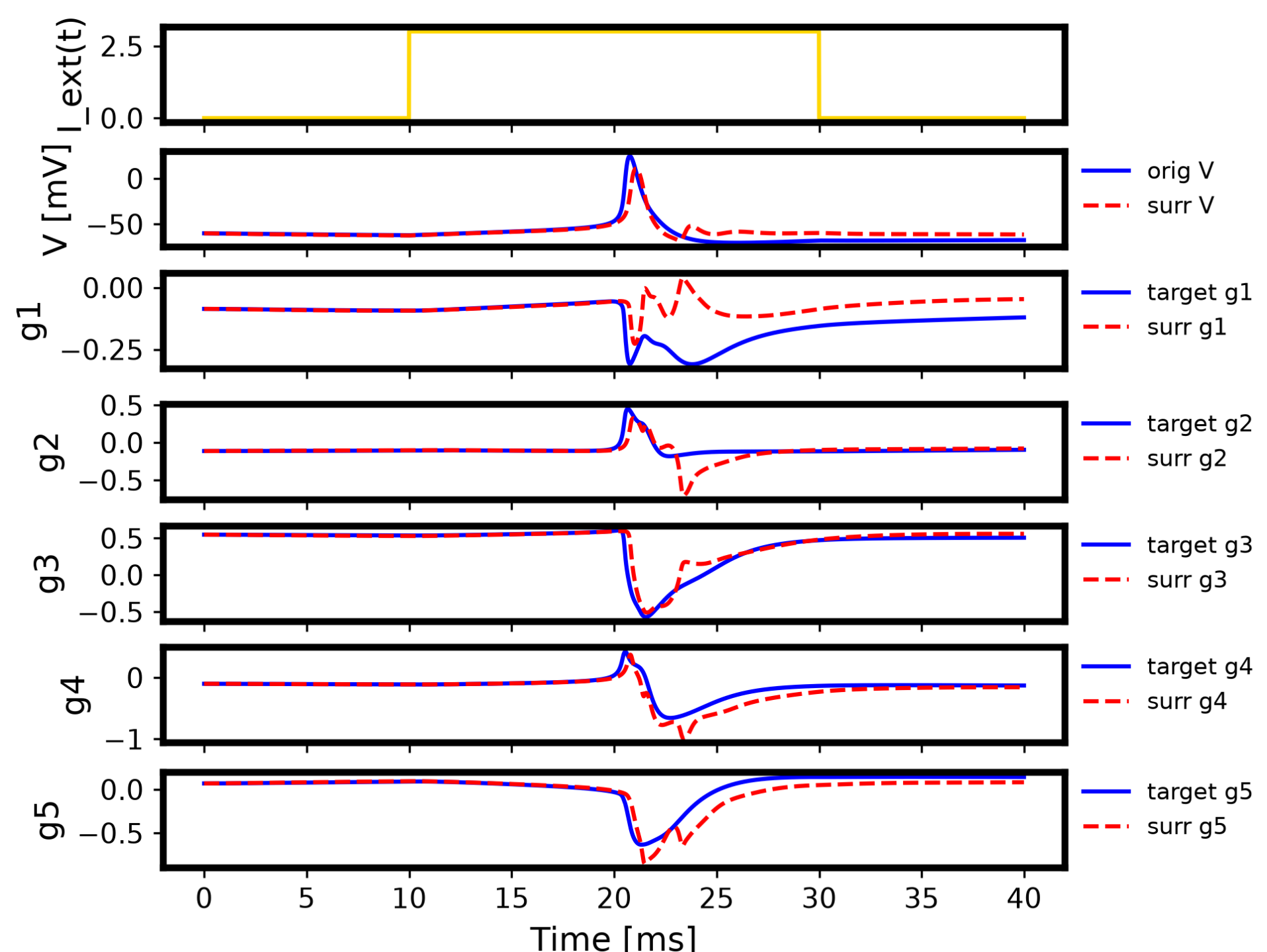


SINDy [3] fits **only $\dot{z}$** ; $\dot{V}$ and Ca^{2+} keep **original physics**. Library is **physics-informed**: gates' own $\alpha(V), \beta(V)$, not a **41-term** generic one.

Each step: $z \rightarrow$ **decode** → gates feed the **unchanged** equivalent-circuit $\dot{V}$; SINDy updates z **in place** of the original gate ODEs.

Results and Discussion

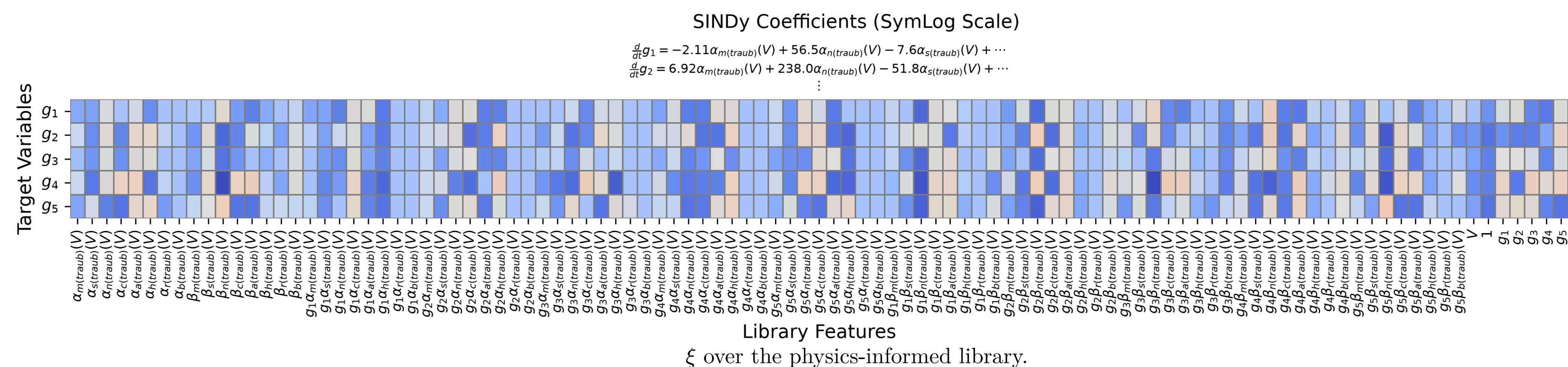
① Single action potential



20 ms, $3 \mu A/cm^2$ step: V and the **5 AE latents** (AE, $n = 5$).

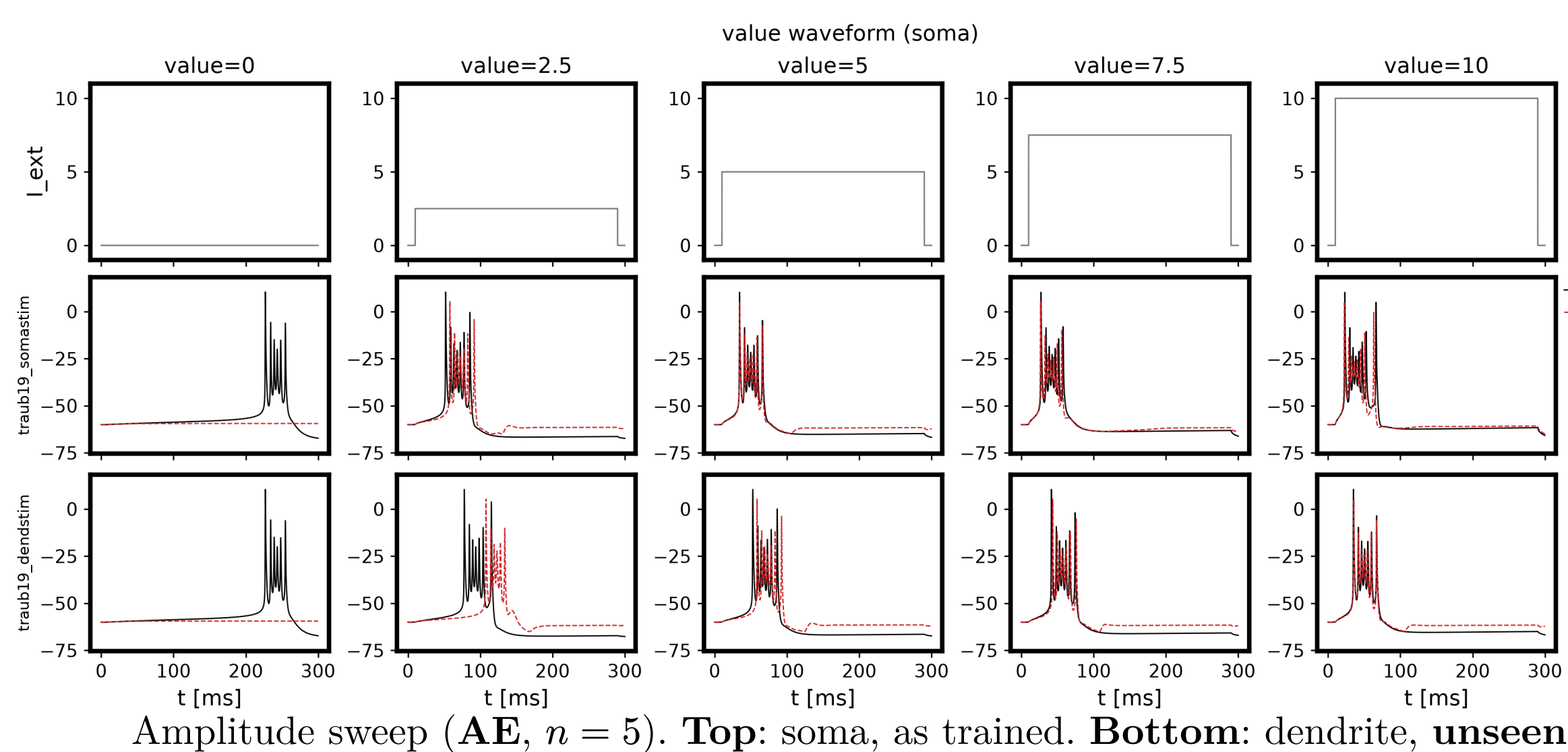
- When it fires is accurate (latency err **0.3 ms**, AHP timing gap **3.1 ms**) — latent dynamics preserve **timing**
 - How **big** is systematically **underestimated** — peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**: error concentrates in the **fast, steep transition**, not the slow relaxation
- suggests the **AE encoding** ($n=5$), not the SINDy library, fails to preserve the **steepest transition**

④ Identified latent equations



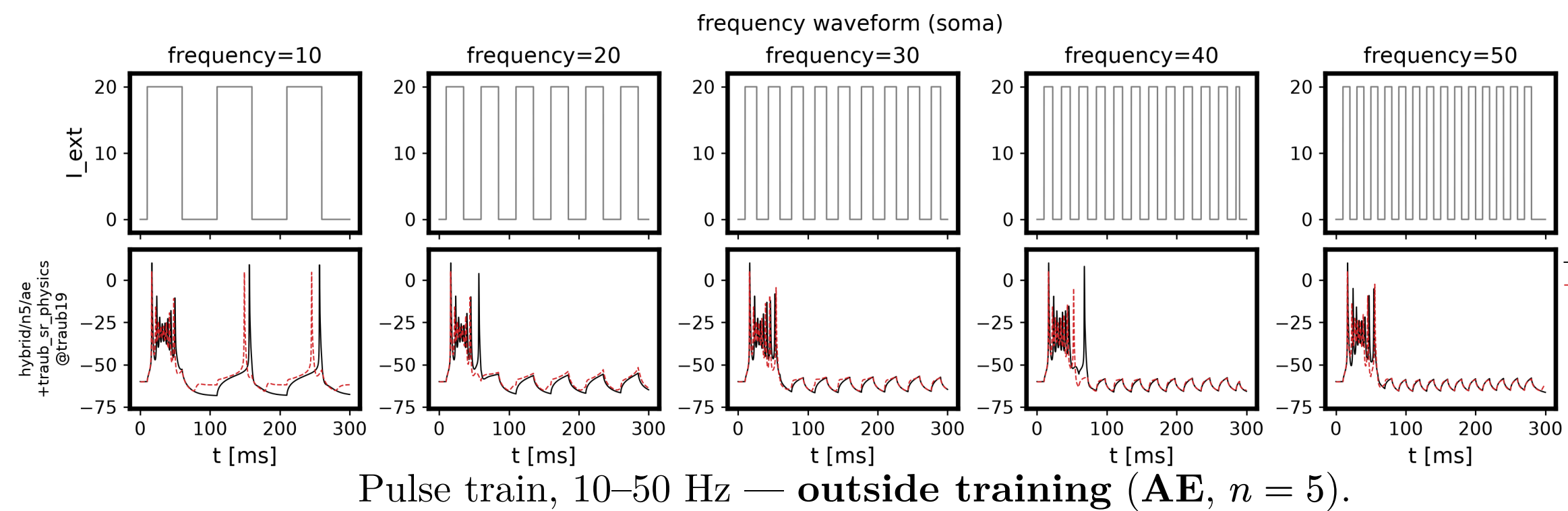
- 79.6% non-zero** — accurate but **not sparse**.

② New stimulus site



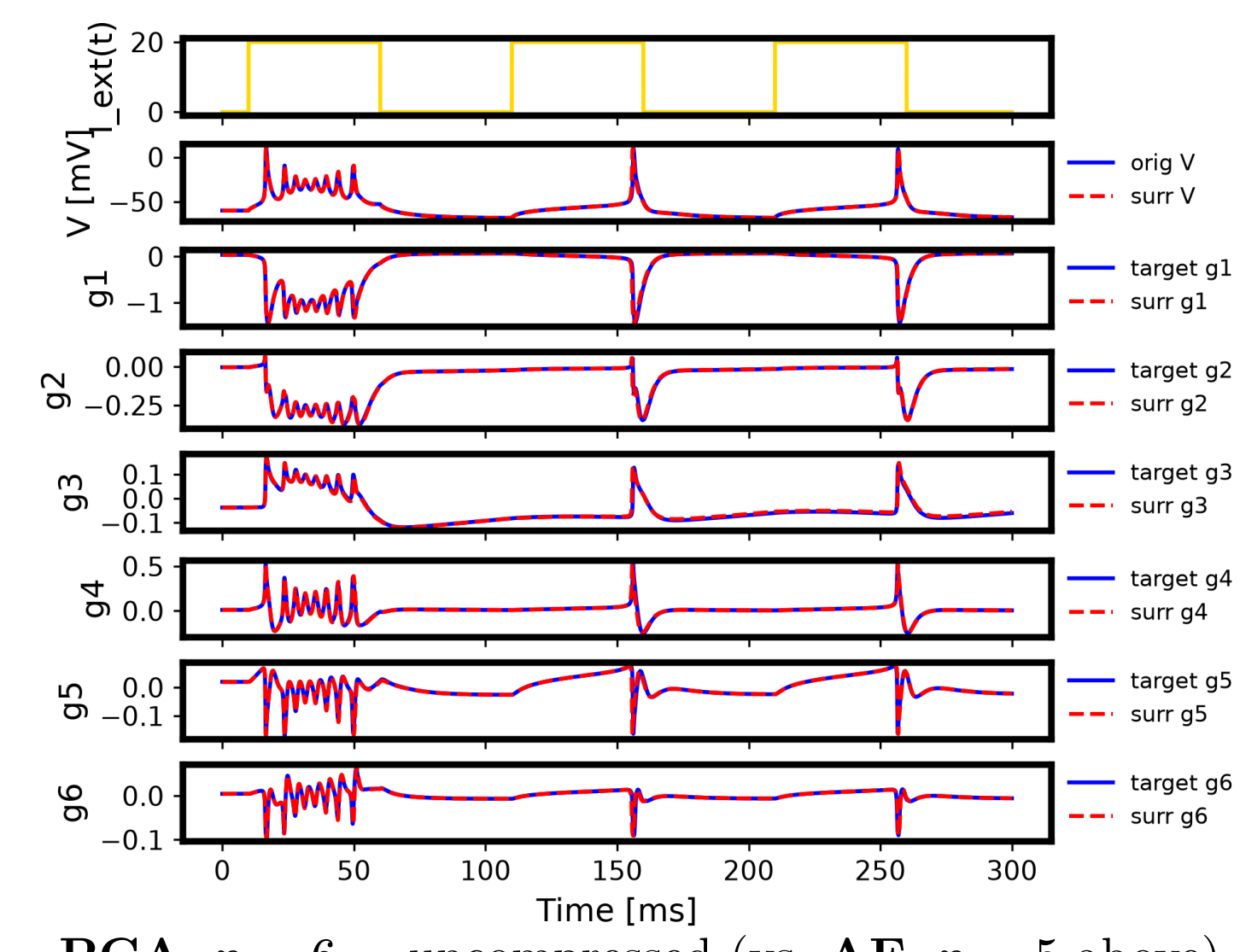
- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

③ Unseen periodic drive



- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

⑤ PCA, no compression



PCA, $n = 6$ = uncompressed (vs. AE, $n = 5$ above).

- Holds **even without compression** → what matters is the **coordinate change itself**, not the compression ratio: core evidence that complex gate dynamics stay **tractable to simple regression**.

Conclusion

- Replaced **all 19 compartments**, no divergence; transfers to **unseen site** • **frequency**.
 - Core finding**: SINDy succeeds **not because we compressed** ($n: 6 \rightarrow 5$) — ⑤ shows the **same result uncompressed** ($n = 6$), so it is the **coordinate change itself** that makes gate dynamics learnable.
 - Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).
- **Future**: sparsify ξ , push n down, test on other channel models.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

- “錐体細胞.” Accessed: Dec. 08, 2025. [Online]. Available: <https://doi.org/10.14931/bsd.2091>
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- K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, “Data-Driven Discovery of Coordinates and Governing Equations,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.